

HW 3

1a

$A \times$
~~9x9~~

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{bmatrix}_{9 \times 1}$$

=

$$\begin{bmatrix} 3 \\ 2 \\ 1 \\ 6 \\ 5 \\ 4 \\ 9 \\ 8 \\ 7 \end{bmatrix}_{9 \times 1}$$

$A =$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}_{9 \times 9}$$

1 4 7
2 5 8
3 6 9

3 2 1
6 5 4
9 8 7

b.

$$A \times \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{bmatrix}_{9 \times 1} = \begin{bmatrix} 3 \\ 6 \\ 9 \\ 2 \\ 5 \\ 8 \\ 1 \\ 4 \\ 7 \end{bmatrix}_{9 \times 1}$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}_{9 \times 9}$$

1 4 7
2 5 8
3 6 9

3 10/3 6
3 5 7
4 17/3 7

d.

$A \times$
 9×9

$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \end{bmatrix}$
 9×1

=

$\begin{bmatrix} 3 \\ 3 \\ 4 \\ 13/3 \\ 5 \\ 17/3 \\ 6 \\ 7 \\ 7 \end{bmatrix}$
 9×1

$A =$

$\begin{bmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix}$
 9×9

- 2a. Standardization is necessary before applying PCA to put everything on the same scale. Values with larger scales will contribute more, which removes the point of PCA.
- b. The first component explains about 36.2% of the variance and the second component explains 19.2% of the variance for a total of about 55.4% of the total variance.
- c. The number of principal components that are needed to explain at least 80% of the variance is 5.
- d. The number of principal components that are needed to explain at least 90% of the variance is 8.
- e. The variables that contribute most strongly to PC1 are flavanols, total phenols, and od280/od315 of diluted wines.
- f. The variables that contribute most strongly to PC2 are color-intensity, alcohol, and proline.
- g. Based on the plot, the classes appear to be clearly separated. The classes form clear separated clusters with very little overlap.

3. The k -value that gives the lowest MSE is $k=5$, with $MSE = 0.008972$. However, the best facial reconstruction comes from $k=30$ with a slightly higher MSE, but a much more detailed face. The choice of k affects denoising based on its size. A small k has a low MSE, but has no key facial details, with a lot of blur. A medium k gets the facial structure and features with some noise. Features like eyes, nose, and mouth are easily seen. A large k will have the most amount of noise as shown by the larger MSE, so there is not a benefit of denoising. When k is too small, there is not enough detail due to the large amount of denoising. When k is too large, there is too much noise and a lack of a purpose for denoising.